

1. In the ordinary operating mode of a 500MW Power plant, as shown in Fig. 1, the expansion valve is closed, and the steam from the boiler passes through the high-pressure turbine to the reheater and into the low-pressure turbine (1 - 2 - 3 - 4 - 5 - 7 - 1). When the high-pressure turbine is out of service, the plant is operated in an emergency mode in which the steam from the boiler is passed through the expansion valve directly into the low-pressure turbine. The modified cycle is now (1 - 2 - 3 - 6 - 7 - 1). Data: $p_1 = 0.04$ bar; $p_2 = 150$ bar; $p_5 = 0.5$ MPa, $x_1 = 0$; $T_3 = 500^\circ\text{C}$; $T_5 = 480^\circ\text{C}$. The isentropic efficiency of the turbines is 0.90, and that of the pump is 0.85. Assume the boiler and reheat unit take heat from a thermal energy reservoir at 600°C and the condenser reject heat to a heat sink of 20°C . Use the data given in the Table 1 for this problem (No data from any steam table is allowed for this problem). [20 Marks]

- Describe the two cycles on $T-s$ and $p-v$ diagrams.
- Calculate the efficiency of each cycle.
- Calculate the mass flow rate of steam, work ratio and specific steam consumption in the ordinary operating mode (i.e., the reheat cycle).
- Find the mean temperature of heat addition for the reheat cycle.
- Find the entropy generation in each component in the ordinary operating mode.
- Calculate the output power in the emergency cycle for the same mass flow rate. Also, calculate the total entropy generation rate.
- Find the irreversibility associated with both the cycles. What are the causes of the irreversibility? Are these cycles internally reversible? If the component isentropic efficiencies are 100% will the cycles be fully reversible? Explain.
- Think of a simple Rankine cycle with same boiler and condenser pressure, boiler exit temperature and component efficiencies as that of the above problem and show in what sense the reheat cycle of the given problem is better than the simple cycle.

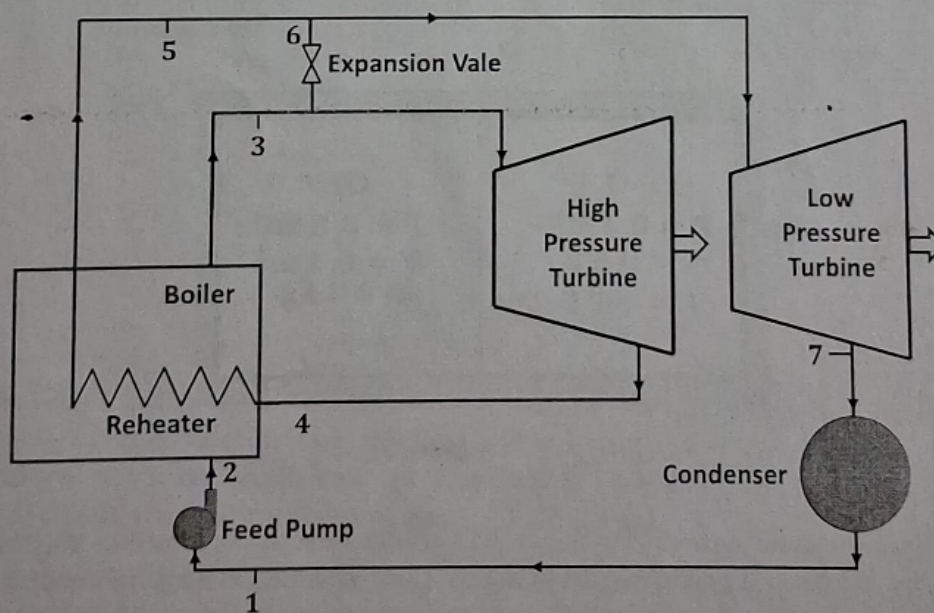


Figure 1

Table 1: Steam Table for problem 1

Pressure bar	Temperature °C	Dryness fraction/quality	Enthalpy kJ/kg	Entropy kJ/kg-K
150	342.3	0	1610	3.684
150	342.3	1	2610	5.309
0.04	29.12	0	121.3	0.4221
0.04	29.12	1	2553	8.473
50	264.1	0	1154	2.92
50	264.1	1	2794	5.973
150	500	—	3309	6.345
150	—	—	136.4	0.4221
50	480	—	3387	6.914
50	—	—	3004	6.345
50	—	—	3309	6.808
0.04	29.12	—	2083	6.914
0.04	29.12	—	1911	6.345
0.04	29.12	—	2051	6.808

$q_{hi-ho} = q_{hi-ho}$
 $q = h_i - h_o$

2. (a) Oxygen (Calorifically perfect gas, $M = 32 \text{ kg/kmol}$, isentropic index i.e., $k = 1.4$) occupies a rigid insulated vessel separated by an adiabatic partition as shown in the Fig. 2. The initial states on both the side of the partition are given in the figure. The partition is broken and equilibrium is reached. Find the final pressure and temperature, entropy change, and entropy generation during the process. Is the process reversible? Explain.
- (b) In the above case if during the mixing process the insulation is removed from the rigid outer wall, what will be the values of the above parameters at the final equilibrium state if the ambient is at 298 K and 101.325 kPa ? [5 Marks]

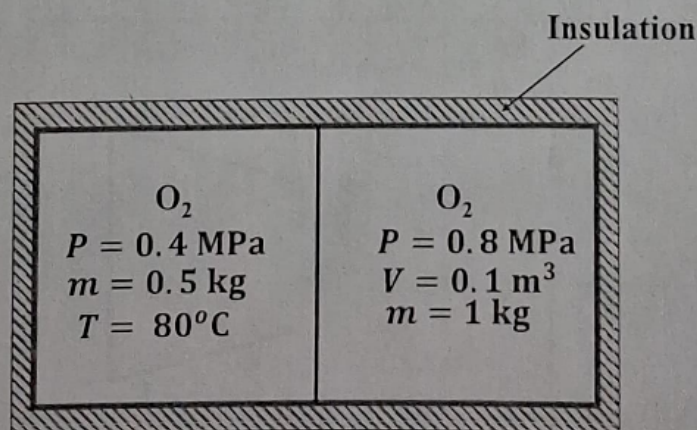


Figure 2

$P V^k = \text{constant}$
 $P \left(\frac{T}{P} \right)^{k-1} = \text{constant}$
 $P^{1-k} T^k = \text{constant}$

3. Steam expands in an imperfectly insulated turbine steadily at a rate of $25,000 \text{ kg/h}$, entering at 6 MPa and 450°C and leaving at 20 kPa as saturated vapor. The power generated by the turbine is 4 MW . The immediate surrounding medium is at a temperature of 25°C . See Fig. 3(a). Neglect changes in the kinetic and potential energies.

$V = m R T$
 $P = \frac{m R T}{V}$

$\frac{P}{T} = \frac{m R}{V}$

$k R$

20

- (a) Show the process in the $(T - s)$ diagram by drawing a neat schematic diagram, showing the vapor dome; saturated, superheated and sub-cooled regions and the critical point; and isobar lines corresponding to inlet and exit conditions. [2 Mark]
- (b) Consider another situation (hereafter referred to as "*Insulated Case*") where the turbine set-up is perfectly insulated, for the given inlet conditions and exit pressure and identical work output. See Fig. 3(b). Determine the isentropic efficiency η_T of the turbine corresponding to the "*Insulated Case*". [2 Marks]
- (c) Determine the net rate of entropy generation for the actual process and compare it with the "*Insulated Case*". [2 Marks]
- (d) For the the actual process and the "*Insulated Case*", does all the entropy production correspond to the steam? Justify your answer. [2 Marks]
- (e) If the specific entropy of the steam at the outlet corresponding to the given exit pressure is identical to that at the inlet, then does it imply that the processes inside the turbine set-up is *internally reversible*? Justify your answer. [2 Marks]
- (f) How do you estimate the maximum theoretical work output of the above turbine for the given inlet conditions and exit pressure? Can it be estimated from the given inlet and outlet thermodynamic conditions? [2 Marks]
- (g) Is it possible to enhance the work output of the above turbine for the given inlet conditions and exit pressure by intentionally heating the system? Justify your answer. [2 Marks]

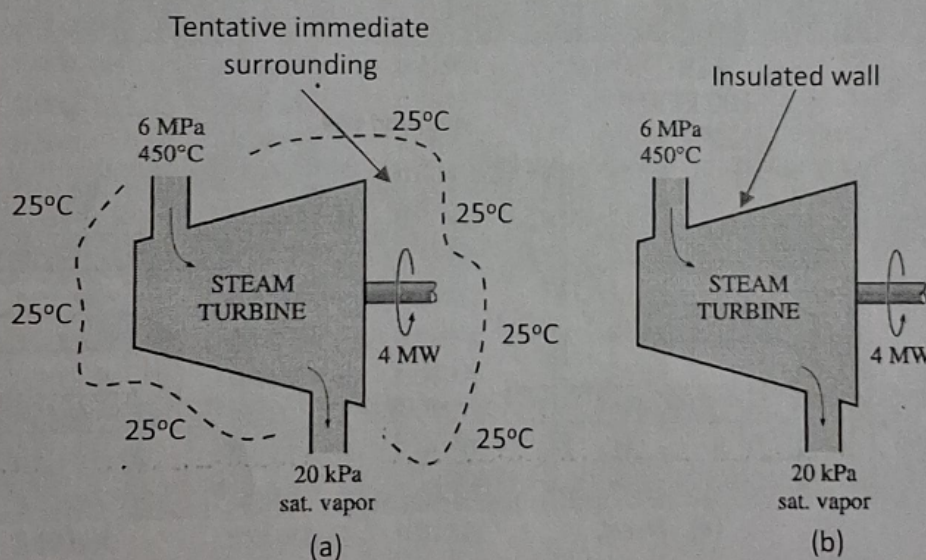


Figure 3

4. The refrigerant-134a ($R-134a$) at 120 kPa with a quality $x = 0.3$ enters steadily the evaporator of a air conditioner at a rate of 2 kg/min and leaves as saturated vapor at the same pressure. The outer surfaces of the air conditioner are insulated as indicated. The inner surface of the pipe carrying $R-134a$ is at thermal equilibrium with the refrigerant.

- (a) Consider the Fig. 4(a), where the evaporator uses air for heating. Air steadily enters the evaporator section at 100 kPa and at an average temperature of 27°C with a volume flow rate of $6 \text{ m}^3/\text{min}$. The average temperature of the outer surface of the evaporator pipe carrying $R-134a$ is roughly at $T_{out} = 32^\circ\text{C}$.

- i. Determine the entropy generation rate for the control volume CV_{1-2} when
 - A. CV_{1-2} contains only $R-134a$, excluding the evaporator pipe. [2 Mark]
 - B. CV_{1-2} contains $R-134a$, including the evaporator pipe. [2 Marks]
- ii. Similarly, determine the entropy generation rate for the control volume CV_{3-4} when
 - A. CV_{3-4} contains only the air, excluding the evaporator pipe. [2 Marks]
 - B. CV_{3-4} contains the air, including the evaporator pipe and excluding $R-134a$. [2 Marks]

Assume air as a calorically perfect ideal gas with Gas Constant $R = 287.05 \text{ J/kg.K}$ and the ratio of specific heats $\gamma = c_p/c_v = 1.4$.

- (b) Consider the Fig. 4(b), where the evaporator is heated by an electric resistance heater, where a current I is flowing steadily over a potential difference of V and the resistance of the part of the circuit inside the evaporator is $\tilde{R}$. The outer surface of the evaporator pipe is insulated for this case.

Determine the entropy generation rate for the control volume CV_{1-2} when

- i. CV_{1-2} contains only $R-134a$, excluding the evaporator pipe and the electric resistance heater. [1 Mark]
- ii. CV_{1-2} contains $R-134a$, including the evaporator pipe, but excluding the electric resistance heater. [1 Mark]
- iii. CV_{1-2} contains only $R-134a$, including the evaporator pipe and part of the the electric resistance heater that is inside the evaporator pipe. [1 Mark]

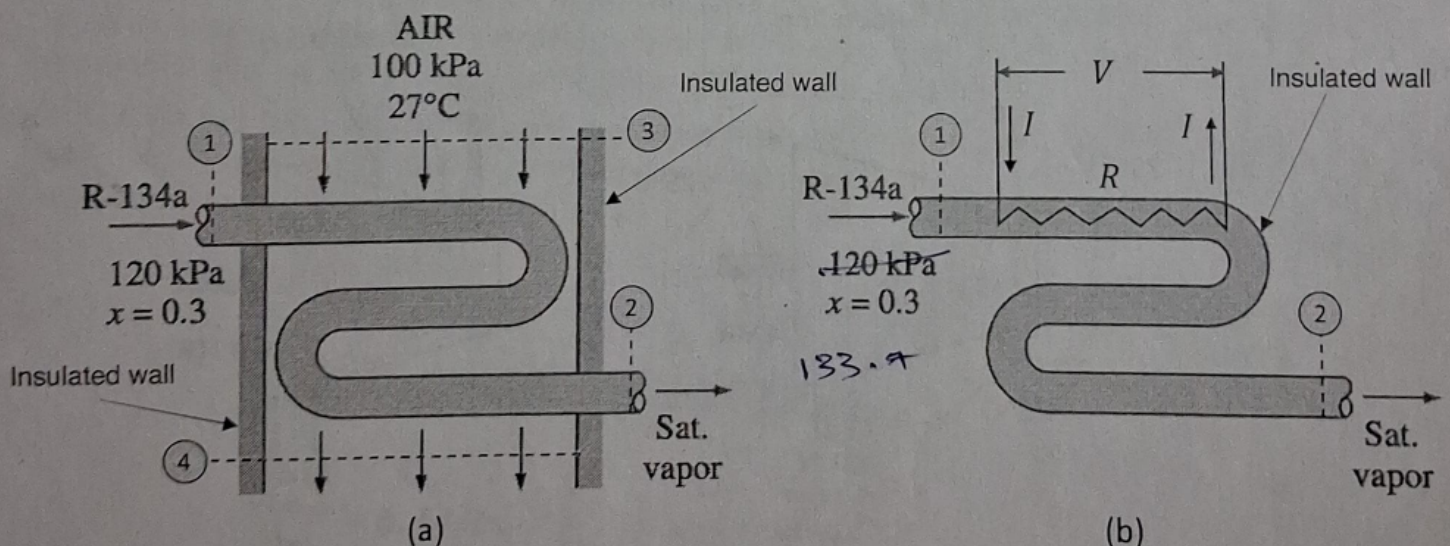


Figure 4